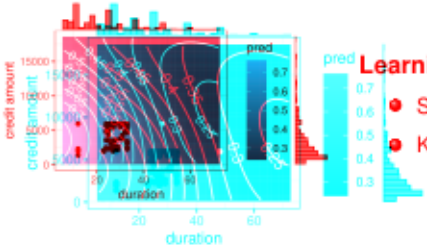
Interpretable Machine Learning

Methods & Discussion of CEs



Learning goals

- See two strategies to generate CEs
- Know problems and limitations of CEs

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OVERVIEW OF METHODS

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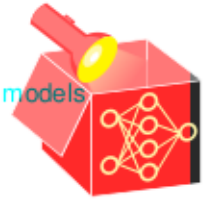
  ⇝ so far, all methods remain in the supervised learning paradigm



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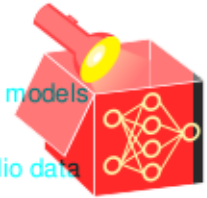

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- **Rashomon Effect:** Many methods return a single counterfactual per run, some multiple counterfactuals, others prioritize CEs or let the user choose

FIRST OPTIMIZATION METHOD

Wachter et al. (2018)

Introduced counterfactual explanations in the context of ML predictions by solving

$$\arg\min_{\mathbf{x}'} \max_{\lambda} \underbrace{\lambda(\hat{f}(\mathbf{x}') - y')^2}_{o_p(\hat{f}(\mathbf{x}'),y')} + \underbrace{\sum_{j=1}^{p} |x'_j - x_j|^2}_{o_d(\mathbf{x}',\mathbf{x})} + \underbrace{\sum_{j=1}^{p} |x'_j - x_j| / MAD_j}_{o_l(\mathbf{x}',\mathbf{x})} \quad (1)$$

MAD_j is the median absolute deviation of feature j. In each iteration, optimizers like Nelder-Mead solve the equation for $\mathbf{x}'$ and then λ is increased until a sufficiently close solution is found

This optimization problem has several shortcomings:

- We do not know how to choose λ a priori
- Due to the maximization of λ, we focus primarily on the minimization of o_p
  $\rightsquigarrow$ only if $\hat{f}(\mathbf{x}') = y'$, we focus on minimizing o_l
- Definition of o_l only covers numerical features
- Other objectives such as sparsity and plausibility of counterfactuals are neglected

MULTI-OBJECTIVE COUNTERFACTUAL EXPLANATIONS

Dandl et al. (2020)

- **Multi-Objective Counterfactual Explanations (MOC):** Instead of collapsing objectives into a single objective, we could optimize all four objectives simultaneously

$$\arg\min_{\mathbf{x}'} \left(o_p(\hat{f}(\mathbf{x}'), y'), o_f(\mathbf{x}', \mathbf{x}), o_s(\mathbf{x}', \mathbf{x}), o_4(\mathbf{x}', \mathbf{X}) \right).$$

- Note that weighting parameters like λ are not necessary anymore
- Uses an adjusted multi-objective genetic algorithm (NSGA-II) to produce a set of diverse counterfactuals for mixed discrete and continuous feature spaces
- Instead of one, MOC returns multiple counterfactuals that represents different trade-offs between the objectives and are constructed to be diverse in feature space

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EXAMPLE: CREDIT DATA

- Model: SVM with RBF kernel
- $\mathbf{x}$: First data point of credit data with $P(y = good) = 0.34$ of being a "good" customer
- Goal: Increase the probability to $[0.5, 1]$
- MOC (with default parameters) found 69 CEs after 200 iterations that met the target
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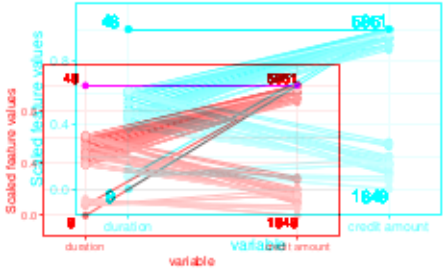
- Parallel plot reveals that all counterfactuals had values equal to or smaller than the values of **x**



Parallel plot: Grey lines show feature values of CEs $\mathbf{x}'$, blue line are values of **x**. Features without proposed changes are omitted. Bold numbers refer to range of numeric features.

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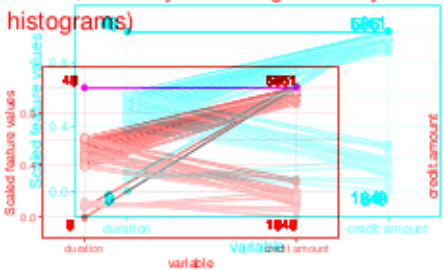
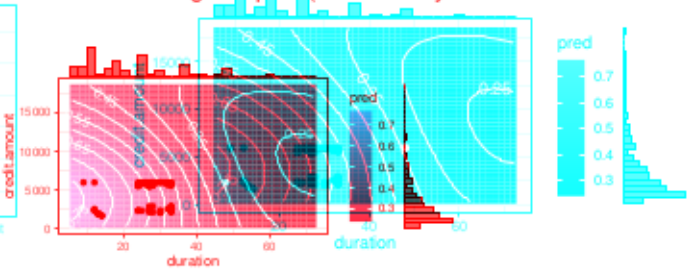
- We can visualize feature changes with a parallel plot and 2-dim surface plot
- Parallel plot reveals that all counterfactuals had values equal to or smaller than the values of **x**
- Surface plot illustrates why these feature changes are recommended
- Counterfactuals in the lower left corner seem to be in a less favorable region far from **x**, but they are in high density areas close to training samples (indicated by histograms)





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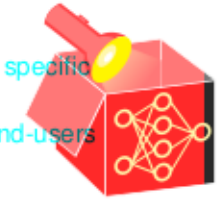
Surface plot: White dot is $\mathbf{x}$, black dots are CEs $\mathbf{x}'$. Histograms show marginal distribution of training data $\mathbf{X}$.

PROBLEMS, PITFALLS, & LIMITATIONS



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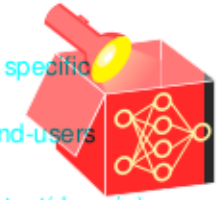

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  ⇝ e.g., L_1 can be reasonable for tabular data but not for image data
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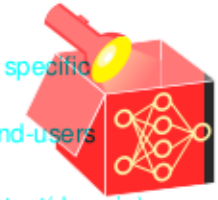

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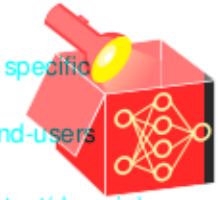

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  ⇝ No perfect solution, depends on end-users computational resources and knowledge

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- **Attacking CEs:** Researchers can create models with great performance, which generate arbitrary explanations specified by the ML developer
  ⇝ how faithful are CEs to the models underlying mechanism?

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